

PA1 | Deep Learning for Space, Time and Graphs

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Repository: <https://github.com/adnan821/DL4STG-PA1.git>

Task 1 - Context dependent forecasting (Responses 1A and 1B)

Setup: 96 observed samples predict the next 48. Stations 1-3 provide training data; Station 4 is held out from fitting. Results below use validation, data/model seeds 0/0. RMSE is in m/s^2 ; MSE is in $(\text{m/s}^2)^2$.

- **Original prediction:** Attention, routed ridge, sensor-specific ridge, shared ridge, from best to worst on established stations. For Station 4: Attention, routed ridge, shared ridge; sensor-specific ridge is unavailable.

Model	Established RMSE / MSE	Station 4 RMSE / MSE
Shared ridge	0.767 / 0.588	0.876 / 0.767
Sensor-specific ridge	0.768 / 0.590	Unavailable
Period-routed ridge	0.166 / 0.028	0.173 / 0.030
Raw Attention	0.434 / 0.188	0.663 / 0.439

Model	Product A	Product B	Product C
Shared ridge	0.805 / 0.909	0.854 / 0.982	0.621 / 0.713
Sensor-specific ridge	0.804 / Unavailable	0.852 / Unavailable	0.630 / Unavailable
Period-routed ridge	0.181 / 0.184	0.164 / 0.182	0.152 / 0.150
Raw Attention	0.436 / 0.710	0.448 / 0.603	0.416 / 0.670

Product RMSE entries show established stations / Station 4.

- **Revised ordering:** Period-routed ridge, Raw Attention, then Shared ridge and Sensor-specific ridge nearly tied. Sensor-specific ridge is unavailable for Station 4. Routed ridge has the lowest error in every product average; its RMSE is 62% below Attention on established stations and 74% below it on Station 4.
- **Why:** each station sees many paces, so identity does not select the right continuation. Routing estimates pace directly. Attention must learn useful comparisons under drift; its flexibility did not win here. Station differences may explain part of its transfer gap, but this run does not isolate the cause.

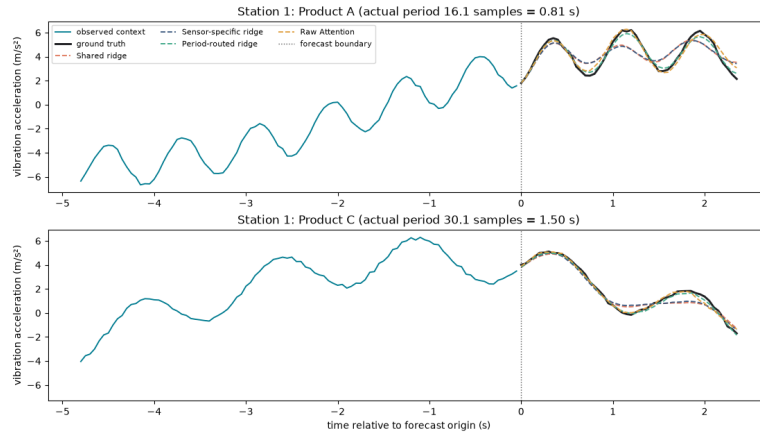


Figure 1. Output 1.2: one station at two paces. Shared ridge flattens later peaks and troughs.

- **Response 1B:** periods 16.1 and 30.1 require about 3 and 1.6 cycles in the horizon. One fixed ridge map compromises in shape (window RMSE 0.779/0.448), while routed ridge reaches 0.232/0.150 and Attention 0.348/0.208.

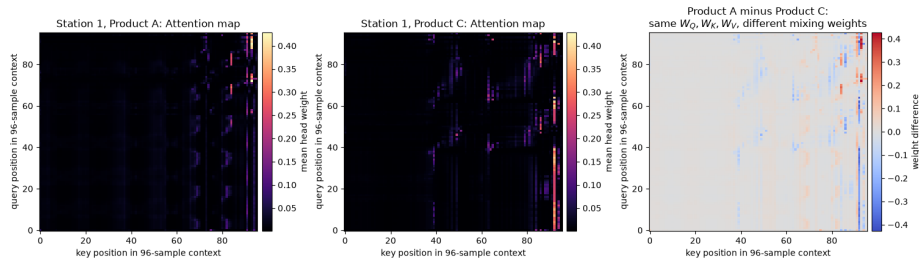


Figure 2. Output 1.3: first layer, averaged across heads Attention maps for the same station.

- Learned Q/K/V projection matrices stay fixed, but their input dependent products create new mixing weights. Mean absolute map difference is 0.013; maximum 0.428. This shows changing mixing, not recovery of the physical period.

Task 1 - Decomposition and delays (Responses 2 and 3)

- **Implementation:** the moving average uses replicated endpoints and returns (input - trend, trend). Delay aggregation reads (t - delay) modulo L. All supplied checks pass in the saved run, including shape and gradient checks.

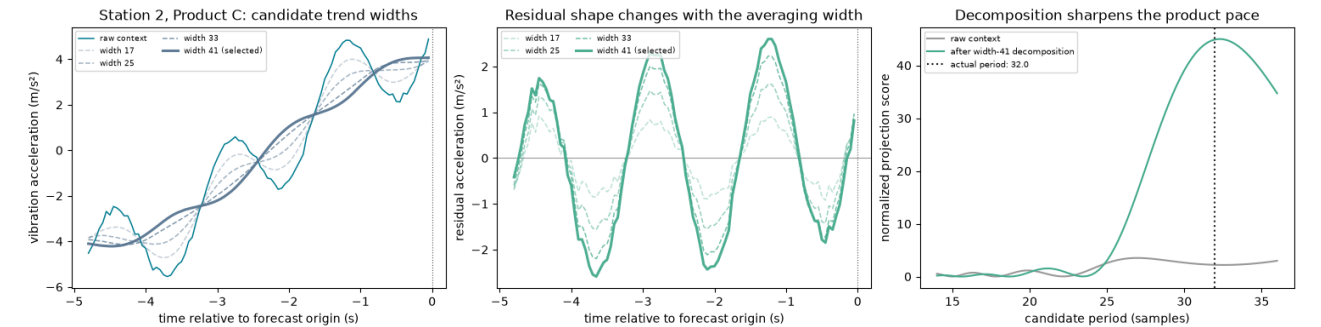


Figure 3. Output 2.1: width 41 exposes repetition more clearly than width 17 in this Product C window.

Representation	Recovery within 1 sample	Median period error
Raw	0.687	0.525
Width 17	0.980	0.157
Width 25	0.989	0.122
Width 33	0.993	0.142
Width 41	0.998	0.216

- **Response 2 - width choice:** 41 has the best training period recovery rate, although 25 has the best median error. The selector uses stride-8 training windows; Output 2.2 uses stride-4 windows. Both favor 41. No validation, test or Station 4 data select the width.
- **Why it helps:** width 41 spans the 14-36 sample vibration periods but is short relative to the 240 sample drift. Averaging suppresses much of the vibration while retaining slower variation. It is not exact: the remainder still contains noise, slow leakage and edge effects, and its amplitude can change.
- **Visual change (Figure 3):** the width 17 trend still follows some vibration. Width 41 gives a smoother trend, a clearer repeating remainder and a sharper period score near 32 samples. A larger remainder does not prove recovery of the true amplitude.
- **Why use the oracle:** it directly tests whether the known synthetic period is easier to recover, without training a neural model for every width. It uses only established station training windows.
- **Forecast evidence (Output 2.3):** adding decomposition changes established RMSE/MSE from 0.434/0.188 to 0.282/0.079, and Station 4 from 0.663/0.439 to 0.432/0.186: about 35% less RMSE on both. All product averages improve, though individual windows can worsen.
- The input trend has a linear forecast head. Decomposition after each residual sublayer reduces slow structure before the next mixer. A sudden level jump is a failure case: smoothing spreads the jump and leaves a false transient. A fitted line or quadratic avoids replicated padding but assumes a simple trend shape and may extrapolate poorly.

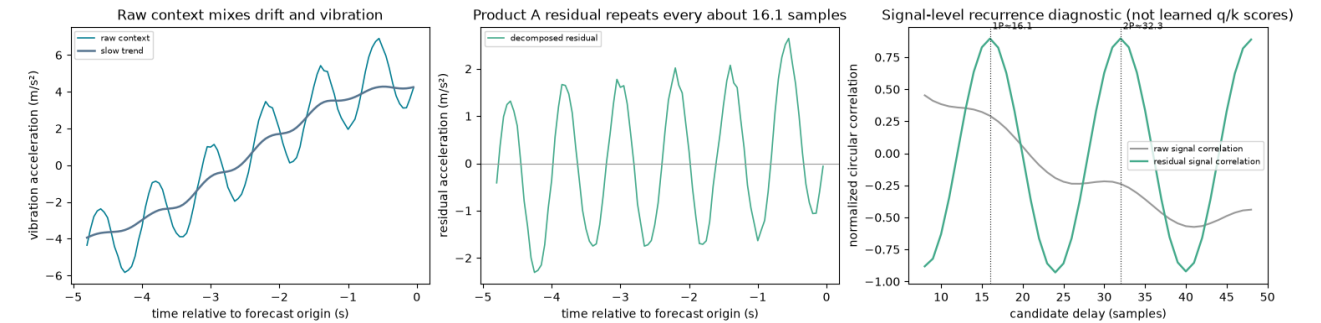


Figure 4. Output 3.2: signal level recurrence for Station 2, Product A. These are not learned q/k scores.

- **Response 3 - checks:** the supplied FFT example gives [4, -4, 4, -4]. Aggregating [10, 20, 30, 40] at delays [1, 3], weights [0.75, 0.25], gives [35, 15, 25, 25]; 35 confirms the direction.
- **Recurrence:** residual correlations peak near P and 2P, at delays 16 (0.896) and 32 (0.897). Delay 48 is also high near 3P, at the band edge. Raw correlations are only 0.29 and -0.24 at 16 and 32; they fall from about 0.45 at delay 8 to -0.57 at 40. The slow ramp obscures the repeating structure.
- **Link to the mixer:** clear recurrence near a few period multiples supports sharing information across recurring delays. It motivates the delay bias, but does not prove which delays the trained mixer selects: that mixer scores learned q/k features, not these raw signal correlations.
- **Limit:** an isolated burst has no stable recurring displacement. Applying the same shifts across positions can spread or misalign it; Attention can assign position specific weights. Fast period drift also weakens shared delays. Circular wrapping can distort even a steady wave when the context does not span whole cycles.

Task 1 - Compare and deploy (Response 4)

- **Original prediction:** decomposition should help most when drift is large, because it separates drift from the mixer. Raw delay mixing was expected to remain affected by drift.

Model	Established RMSE / MSE	Station 4 RMSE / MSE	Parameters	Fit (s)
Shared ridge	0.767 / 0.588	0.876 / 0.767	4,608	0.003
Sensor-specific ridge	0.768 / 0.590	Unavailable	13,824	0.006
Period-routed ridge	0.166 / 0.028	0.173 / 0.030	59,904	0.013
Raw Attention	0.434 / 0.188	0.663 / 0.439	368,368	176.9
Raw delay	0.221 / 0.049	0.358 / 0.128	368,370	259.8
Attention + decomposition	0.282 / 0.079	0.432 / 0.186	373,024	204.5
Autoformer-inspired	0.198 / 0.039	0.246 / 0.060	373,026	273.4

- **Switches:** decomposition alone reduces MSE by 58% on both populations; delay mixing alone by 74%/71%. Adding decomposition to delay mixing reduces MSE by another 20%/53%. Percentages use rounded values.
- **Revisiting the prediction:** delay mixing is the stronger single switch overall. The earlier assumption about drift was too strong because the mixer scores learned features. Overall averages do not directly test the prediction for contexts with especially large drift.
- **Interaction:** individual established gains sum to 0.248 versus 0.149 combined; Station 4 gains sum to 0.564 versus 0.379 combined. Gains are smaller than additive on the MSE scale. This is consistent with overlap, but the zero loss floor and single seed prevent a causal conclusion. The interaction was not stated in the original prediction.
- **Exceptions (Output 4.1):** raw delay beats the combined model on established Product A (0.211 versus 0.225 RMSE). In the largest gain over Attention, Station 4/Product C improves 1.450 to 0.400; raw delay is better still at 0.298. The largest loss is Station 1/Product A, 0.200 to 0.310.

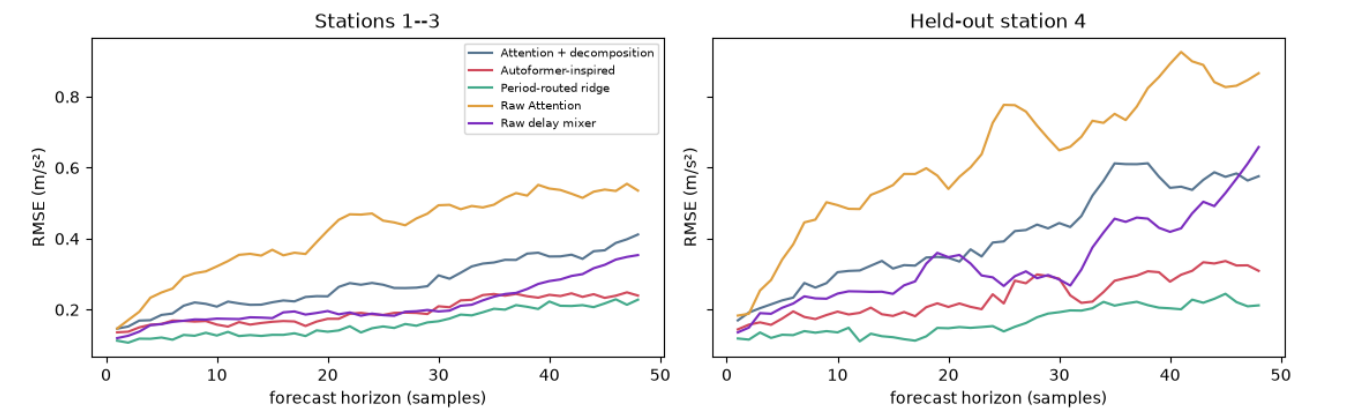


Figure 5. Output 4.2: errors generally rise with horizon, especially for Raw Attention on Station 4.

- At step 48, approximate established/Station 4 RMSEs are 0.54/0.87 for Attention, 0.41/0.57 with decomposition, 0.35/0.66 for raw delay, 0.25/0.31 combined, and 0.23/0.22 for routed ridge. These are direct forecasts, not recursive error feedback.
- **Deployment:** choose period-routed ridge for both populations using validation. It has the lowest MSE, about six times fewer parameters than the best neural model, and much lower fitting cost. Output 4.3 gives test MSE/RMSE 0.026/0.161 established and 0.033/0.183 on Station 4. Fit times are not inference speed measurements.

Interpretation (under 200 words). Explicit pace routing is the best choice for this benchmark. Both neural upgrades help in aggregate, and delay mixing is the stronger single switch. Their combination is best among neural models overall, but not for every product or window. Reductions smaller than their sum suggest overlapping benefits, without proving a shared mechanism. The results favor useful temporal assumptions over extra flexibility here; they do not establish which model is safest at unseen periods.

Task 2 - Aim and evidence

- **Aim:** forecast indices 43,657-43,824 using Autoformer. This section follows the saved experiment logs and forecasts in the newer Task 2 implementation, rather than the earlier notebook scores.
- **Validation:** the main comparison uses twenty consecutive 168-step blocks after a training cutoff of 40,296 observations. Scaling is fitted on training data only. RMSE below means the average of the twenty block RMSEs, not pooled RMSE. Additional checks use earlier cutoffs of 36,936 and 33,576.

Task 2 - How the model improved (Section 2.8)

- **Start simple:** I used a compact Autoformer with moving average decomposition and FFT delay mixing. The first sweep, already using raw covariates, gave 66.2 RMSE on average over two seeds. Making the model larger or heavily regularized did not reliably help.
- **Give covariates a direct route:** I added a linear covariate correction to the decoder's initial trend. Combined with square root target training, this reduced the two-seed mean to 61.4. The trend correction alone scored 67.8, so the gain was not from that change alone.
- **Tune context and average seeds:** eight epochs with context 336 gave a three-seed ensemble RMSE of 58.78. Context 504 became the preferred recipe; its corrected rerun scored 55.91. Longer context 720, raw target training and alternative memory alignment did not improve the selected recipe.
- **Extract more from the optional file:** I expanded the ten covariates to 27 features and replaced the linear trend correction with a small nonlinear network. The three-seed ensemble improved from 55.91 to 49.44 RMSE, an 11.6% reduction. It also improved at both earlier cutoffs, reaching 40.73 and 69.14, versus baseline values of about 43.7 and 75.6.
- **Normalization, not RevIN:** the saved final implementation uses global means and standard deviations fitted on training data, with a square root target transform. It does not use reversible per-window normalization (RevIN). The verified gains above should therefore be attributed to the tested recipe, not to RevIN.

What the optional file contributed

- **Where it enters:** optional_external_data.csv supplies ten aligned variables for both history and the 168 future positions. Past covariates join the target in the encoder; label and future covariates enter the decoder and its trend correction. No future target values are supplied.
- **What I added:** log1p transforms for D, E and F; the A minus B contrast; and trailing 6/24-step means and lag differences for the first four transformed variables. These give 27 external features. Each uses only external values at the same or earlier index. The trend network is 27-32-32-1 with GELU activations.

L504 input ablation	Mean seed RMSE	Ensemble RMSE
No external inputs	86.74	86.27
Historical external inputs only	89.35	85.51
Historical and future external inputs	59.67	55.91

- **Main finding:** with the same context, square root training, eight epochs and three seeds, allowing future external inputs reduced ensemble RMSE by 30.36 points (35.2%) compared with no external data. Past inputs alone gave little benefit. Knowing the external conditions over the forecast horizon was the important difference.
- **Leaderboard versus validation:** the project records report that the richer covariate model reduced hidden-test RMSE from 80.60 to 78.31, and score from 80.6242 to 78.3292. This was a smaller gain than on validation; validation improvements do not guarantee the same improvement on one hidden block.

Task 2 - Data, final model and ensemble

- **Data:** 43,656 observed nonnegative values and 168 hidden targets. The target is skewed (median 73, mean 98.2, maximum 994), motivating comparisons of raw, log and square root training. The time interval and physical feature meanings are undisclosed.

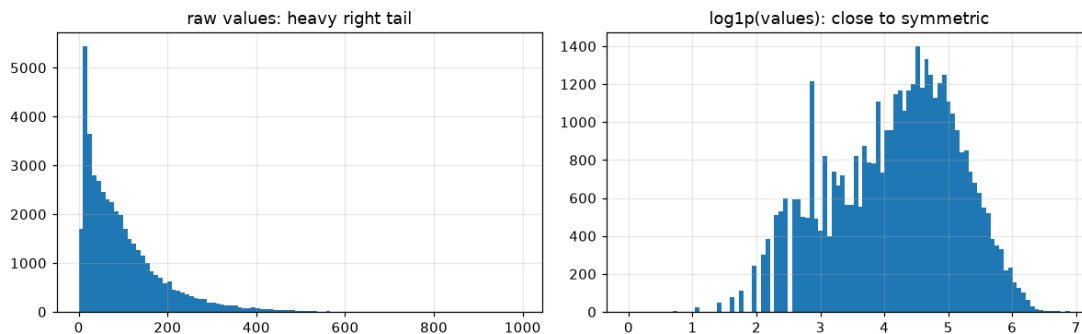


Figure 6. Original Task 2 notebook, Section 2.2: raw and log transformed target distributions. This describes the data; the selected model uses a square root transform.

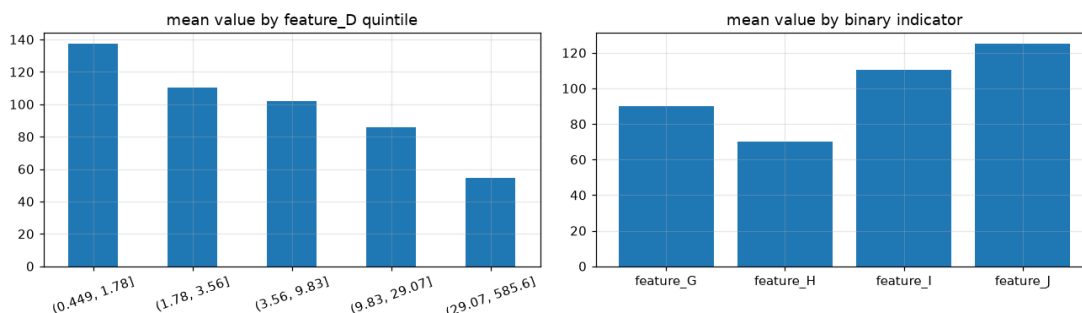


Figure 7. Original Task 2 notebook, Section 2.6: target means vary across external input groups. The matched ablation above, rather than this association alone, measures predictive benefit.

- **Core model:** context 504, label 168, horizon 168; width 32, four heads, feedforward width 64, one encoder and one decoder layer. Width-25 moving averages separate streams. FFT scoring selects $\text{ceil}(\log L)$ delays per example. Decomposition and delay mixing remain in the prediction path after adding the covariate network.
- **Training:** AdamW with learning rate 0.001 and a OneCycle schedule, batch 64, stride 1, eight fixed epochs. MSE is applied in standardized square root target space. Predictions are restored to original units, with negative square root predictions clamped before squaring.
- **Ensemble:** train the same selected architecture with five seeds (0-4) on all 43,656 observed targets. Each member produces 168 predictions. Average the five forecasts at each horizon in original target units. This reduces dependence on one initialization without combining different model families.
- **Cost:** the final covariate-network model has 28,482 parameters per member: $P = 142,410$ for five members, $E = 40$ for five eight-epoch fits. The three-seed validation results above are not five-seed ensemble scores.
- **Evidence source:** numerical comparisons use the newer saved Task 2 logs and final-epoch validation forecasts. The two figures are retained from the original notebook because they describe the same input data. No new training was required to calculate these comparisons.

Optional additional run

- I am also running a revised notebook on another computer. In that notebook, I corrected scaling to use training data only, added a matched time only control, and separated historical from future external inputs. I also added validation of ensembles with 1, 3 and 5 seeds, cost based selection within 1% of the best validation RMSE, and clearer parameter and epoch accounting.
- This run is still in progress, so I do not claim an RMSE improvement from it. I will upload it as optional supporting work; it may not count toward the assignment score. The completed results reported above remain separate from this run.

Task 2 - Short reflections (Section 2.9)

- **Q2.1 - Delays versus positions.** Stable repeating signals favor a few shared delays: many position pairs reuse the same relation. Isolated events or position specific dependencies favor Attention. No universal strict ranking follows from architecture alone. Rapid period drift spreads delay peaks; Attention can adapt locally but must learn the changing relation.
- **Q2.2 - Moving average assumptions.** The trend should change slowly across the averaging window. Replicated edges bias endpoints, curvature is smoothed, and a sharp step becomes a ramp. A line or quadratic fit assumes a simple window wide trend instead; it can miss complex drift and extrapolate badly.
- **Q2.3 - Repeated decomposition.** Later layers can reintroduce slow structure, so repeated splitting keeps it out of subsequent seasonal mixers. It can hurt when a slow component carries useful seasonal information or when repeated filtering compounds edge distortion.
- **Q2.4 - Number of delays.** Small k focuses the mixer but can miss useful periods; large k includes more alternatives but also weak or conflicting shifts. The log L rule is a practical budget, not a universal optimum. Several real periodicities may need more delays, selected using validation.
- **Q2.5 - Direct versus recursive forecasts.** Direct prediction avoids feeding forecast mistakes back as inputs, but must learn all horizons jointly. Recursive prediction can exploit a good short step transition rule, but teacher forced training creates exposure bias and shared early errors propagate. Longer, noisy horizons often favor direct prediction; stable known dynamics can favor recursion.
- **Q2.6 - Window normalization.** Normalized inputs lose absolute level and scale unless these are retained as separate inputs. Restoring them later only permits the restoration rule to use them. This helps offset/scale shifts. It fails when identical normalized shapes have different level dependent dynamics, such as a physical threshold triggering a different future.
- **Q2.7 - Input availability.** Historical inputs inform the encoder; known future inputs can condition each decoder step. An independent future control command is useful if supplied but cannot be inferred from its history. A static site attribute is equally available and useful in both cases. Availability changes placement as well as potential benefit.
- **Q2.8 - Missing calendar.** Without dates, weekday, holiday and daylight saving effects cannot be identified directly. ACF, spectral peaks and index based Fourier features can recover stable cycles, but require regular sampling and periods that persist. A 24-step peak does not by itself mean one day.
- **Q2.9 - Classical versus deep models.** Short, noisy series with stable trend/seasonality often favor simpler models. More data, changing relationships and informative inputs may justify a larger model. Use learning curves and chronological tests against simple baselines; there is no universal crossover. Here, the submitted architecture must still be Autoformer.

References and AI-use disclosure

Sources: AI651 PA1 handout; supplied Task 1 harness; Wu et al. (2021), *Autoformer*, Sections 3.1-3.2; official implementation: <https://github.com/thuml/Autoformer>. Task 2 is adapted from these mechanisms and Task 1, with per example delay selection and added input paths.

AI assistance: I used Codex for code review and help with LaTeX formatting for this report.